



Lakshya JEE 2.0 (2024)

Definite Integration

DPP-05

1. $f(x) = \begin{vmatrix} e^{x^2} & \sec x & \sin x \\ \cos x & \tan^2 x & x \\ 3 & 4 & x + \tan x \end{vmatrix}$ then value of

$$\int_{-\pi/2}^{\pi/2} (x^2 + 1)(f(x) + f''(x))dx \text{ is}$$

- (1) 0 (2) 1
(3) -1 (4) 2

2. $\lim_{n \rightarrow \infty} \sum_{r=1}^{4n} \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r} + 4\sqrt{n})^2}$ is equal to

- (1) $\frac{1}{35}$ (2) $\frac{1}{14}$
(3) $\frac{1}{10}$ (4) $\frac{1}{5}$

3. Let $S_n = \frac{n}{(n+1)(n+2)} + \frac{n}{(n+2)(n+4)} + \frac{n}{(n+3)(n+6)} + \dots + \frac{1}{6n}$, then $\lim_{n \rightarrow \infty} S_n$ has the value of equal to:

- (1) $\ln \frac{3}{2}$ (2) $\ln \frac{9}{2}$
(3) $2 \ln \frac{3}{2}$ (4) $\frac{1}{2} \ln \frac{3}{2}$

4. Let $\alpha > 1$ and $\beta > -1$, then the value of

$$\lim_{n \rightarrow \infty} n^{\beta-\alpha} \left(\frac{1^\alpha + 2^\alpha + \dots + n^\alpha}{1^\beta + 2^\beta + \dots + n^\beta} \right) \text{ is :}$$

- (1) $\frac{\beta+1}{\alpha+1}$ (2) $\frac{\alpha+1}{\beta+1}$
(3) $\frac{\alpha+2}{\beta+2}$ (4) $\frac{\beta+2}{\alpha+2}$

5. The value of

$$\lim_{n \rightarrow \infty} \frac{(1^2 + 2^2 + \dots + n^2)(1^3 + 2^3 + \dots + n^3)(1^4 + 2^4 + \dots + n^4)}{(1^5 + 2^5 + \dots + n^5)^2}$$

is equals to:

- (1) $\frac{4}{5}$ (2) $\frac{3}{5}$
(3) $\frac{2}{5}$ (4) $\frac{1}{5}$

6. The value of

$$\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{1/n} \text{ is}$$

- (1) $\frac{e^{\pi/2}}{2e^2}$ (2) $2e^2 e^{\pi/2}$
(3) $\frac{2}{e^2} e^{\pi/2}$ (4) 0

7. If $L = \lim_{n \rightarrow \infty} \frac{1}{n^4} \prod_{i=1}^{2n} (n^2 + i^2)^{\frac{1}{n}}$, then $\ln L$ is equal to :

- (1) $\ln 2 + \frac{\pi}{2} - 2$
(2) $2 \tan^{-1} 2 - 4 + 2 \ln 5$
(3) $2 \tan^{-1} 2 + 4 + 2 \ln 5$
(4) $2 \ln 5 - 4 - 2 \tan^{-1} 2$

8. $\lim_{n \rightarrow \infty} \left[(1+n)^{\frac{1}{n}} \left(1 + \frac{n}{2}\right)^{\frac{2}{n}} \left(1 + \frac{n}{3}\right)^{\frac{3}{n}} \dots 2 \right]^{\frac{1}{n}}$ is equal to

- (1) e (2) $e^{\frac{1}{2}}$
(3) $e^{\frac{1}{4}}$ (4) $e^{\frac{-1}{2}}$

Evaluate the following limits

9. $\lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2 + n}} + \frac{1}{\sqrt{n^2 + 2n}} + \dots + \frac{1}{\sqrt{n^2 + n^2}} \right]$



$$10. \quad \lim_{n \rightarrow \infty} \left[\frac{1}{1+n} + \frac{1}{2+n} + \frac{1}{3+n} + \dots + \frac{1}{5n} \right]$$

$$11. \quad \lim_{n \rightarrow \infty} \frac{1}{n^2} \left[\sin^3 \frac{\pi}{4n} + 2 \sin^3 \frac{2\pi}{4n} + \dots + n \sin^3 \frac{n\pi}{4n} \right]$$

$$12. \quad \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{\sqrt{n^2 - r^2}}$$

$$13. \quad \lim_{n \rightarrow \infty} \frac{3}{n} \left[1 + \sqrt{\frac{n}{n+3}} + \sqrt{\frac{n}{n+6}} + \sqrt{\frac{n}{n+9}} + \dots + \sqrt{\frac{n}{n+3(n-1)}} \right]$$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)
2. (3)
3. (1)
4. (1)
5. (2)
6. (3)
7. (2)
8. (3)

9. $2(\sqrt{2} - 1)$
10. $\log_e 5$
11. $\frac{\sqrt{2}}{9\pi^2} (52 - 15\pi)$
12. $\frac{\pi}{2}$
13. 2



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8eid80if>